CODEFORCES
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Warmup Contest

A. Parking Fee

time limit per test: 1 second
memory limit per test: 256 megabytes

A mall parking lot charges cars for the time they stay.

The first 2 hours are free. After that, every started hour costs 20 rupees, but no car ever pays more than 100 rupees. For example, a car that stays 2 hours and 1 minute pays 20 rupees, and so does a car that stays exactly 3 hours.

A car enters at time h_1:m_1 and leaves at time h_2:m_2 on the same day. Find how much it pays.

Input

The first line contains two integers h_1 and m_1 ($0 \le h_1 \le 23, 0 \le m_1 \le 59$): the time the car enters.

The second line contains two integers h_2 and m_2 ($0 \le h_2 \le 23, 0 \le m_2 \le 59$): the time the car leaves. The car does not leave before it enters.

Output

Print the amount the car pays, in rupees.

Examples

input

```
10 15
14 40
```

output

```
60
```

input

```
9 0
18 30
```

output

```
100
```

Note

In the first example, the car stays from 10:15 to 14:40, which is 4 hours and 25 minutes. The first 2 hours are free. The remaining 2 hours and 25 minutes make 3 started hours, so the car pays $3 \times 20 = 60$ rupees.

In the second example, the car stays 9 hours and 30 minutes. After the free 2 hours, 8 hours are started, which would cost 160 rupees, but no car pays more than 100 rupees.

B. Walking the Wire

time limit per test: 1 second
memory limit per test: 256 megabytes

Brand New Day: nobody remembers Peter Parker, and he is testing a new line of web stretched between two rooftops with MJ.

The line is made of n knots in a row, the i-th tied in colour s_i. Peter starts on the first knot and MJ on the last. Every second, both of them call out the colour of the knot they are standing on, then each steps one knot toward the other.

They want to **meet**: end up standing on the same knot, having called out the same colour at every call along the way. If they ever call different colours, or step past each other without ever sharing a knot, they fail.

Determine whether they meet.

Input

The first line contains one integer t ($1 \le t \le 100$): the number of test cases.

Each test case is one line with a string s of lowercase English letters ($1 \le |s| \le 10^5$). The total length of all strings does not exceed 10^5.

Output

For each test case, print `YES` if they meet, and `NO` otherwise.

Examples

input

```
3
abcba
abca
z
```

output

```
YES
NO
YES
```

input

```
1
ab
```

output

```
NO
```

Note

In the first test case of the first example they call `a`, then `b`, and then both stand on the middle knot, coloured `c`. In the second, Peter calls `b` while MJ calls `c` at the second call.

C. Shrinking Number

time limit per test: 1 second
memory limit per test: 256 megabytes

Take a positive integer n. In one step, n is replaced by the product of its digits. For example, in one step 77 becomes $7 \times 7 = 49$.

The steps are repeated until n has only one digit. Find the number of steps.

Input

The first and only line contains an integer n ($1 \le n \le 10^{18}$).

Output

Print the number of steps needed until n has only one digit.

Examples

input

```
77
```

output

```
4
```

input

```
7
```

output

```
0
```

Note

In the first example, $77 \to 49 \to 36 \to 18 \to 8$: $7 \times 7 = 49$, $4 \times 9 = 36$, $3 \times 6 = 18$ and $1 \times 8 = 8$. After 4 steps only one digit is left.

In the second example, 7 already has only one digit, so no steps are needed.

D. Mirror Primes

time limit per test: 1 second
memory limit per test: 256 megabytes

A number is a mirror prime if it is prime and it is still prime when its digits are written in reverse order.

For example, 13 is a mirror prime, because both 13 and 31 are prime. 23 is not a mirror prime, because 32 is not prime.

Given two integers L and R, count the mirror primes between L and R, inclusive.

Input

The first and only line contains two integers L and R ($1 \le L \le R \le 1000$).

Output

Print the number of mirror primes x with $L \le x \le R$.

Examples

input

```
10 40
```

output

```
5
```

input

```
1 10
```

output

```
4
```

Note

In the first example, the primes between 10 and 40 are $11, 13, 17, 19, 23, 29, 31$ and 37. Their reverses are $11, 31, 71, 91, 32, 92, 13$ and 73. $91 = 7 \times 13$, and 32 and 92 are even, so $19, 23$ and 29 are not mirror primes. The mirror primes are $11, 13, 17, 31$ and 37, so the answer is 5.

In the second example, the mirror primes are 2, 3, 5 and 7.

E. Sales Streak

time limit per test: 1 second
memory limit per test: 256 megabytes

A shop records its sales for n days in a row. On day i, it sells a_i items.

A streak is a run of one or more consecutive days in which the sales never drop: every day after the first sells at least as many items as the day before it.

Find the longest streak. If there are several longest streaks, choose the one that starts first.

Input

The first line contains an integer n ($1 \le n \le 1000$).

The second line contains n integers $a_1, a_2, \ldots, a_n$ ($0 \le a_i \le 10^9$).

Output

Print two integers: the length of the longest streak and the day it starts.

Examples

input

```
7
5 3 4 4 6 2 7
```

output

```
4 2
```

input

```
6
1 3 2 5 4 6
```

output

```
2 1
```

Note

In the first example, days 2 to 5 sell 3, 4, 4 and 6 items. The sales never drop, so this is a streak of 4 days starting on day 2. Every other streak is shorter. For example, days 6 and 7 (2 and 7 items) form a streak of only 2 days.

In the second example, days 1 and 2, days 3 and 4, and days 5 and 6 form streaks of 2 days, and no streak is longer. The first of them starts on day 1.

F. Cinema Seats

time limit per test: 1 second
memory limit per test: 256 megabytes

A cinema hall has n rows with m seats in each row. Some of the seats are already booked.

A group of k friends wants to watch a movie together, so they want k free seats next to each other in the same row.

Count the number of ways to choose such a block of seats. Two ways are different if they use different seats.

Input

The first line contains three integers n, m and k ($1 \le n, m \le 100$, $1 \le k \le m$).

Each of the next n lines contains a string of m characters describing one row of the hall: `.` is a free seat and `#` is a booked seat.

Output

Print the number of ways the friends can choose k free seats next to each other in the same row.

Examples

input

```
3 5 2
..#..
.....
##.#.
```

output

```
6
```

input

```
4 3 3
#.#
#.#
#.#
...
```

output

```
1
```

Note

In the first example, in the first row the friends can take seats 1 and 2, or seats 4 and 5. In the second row, they can take seats 1 and 2, 2 and 3, 3 and 4, or 4 and 5. The third row has no two free seats next to each other. So there are $2 + 4 + 0 = 6$ ways.

In the second example, only the last row has 3 free seats next to each other. The free seats of the middle column are in different rows.

G. Spell Runes

time limit per test: 1 second
memory limit per test: 256 megabytes

To seal the crack in the sky, Doctor Strange carves a rune pattern of n rows onto the floor of the Sanctum.

The runes are carved in the order A, 0, B, 1, C, 2, D, 3, ...: letters and digits alternate. The letters run from A to Z and then start again from A, and the digits run from 0 to 9 and then start again from 0.

Every row is $2n$ characters wide. Row i takes the next i runes for its left end and the next i runes for its right end, and is filled with spaces in between. Rows are carved from top to bottom, and the left end of a row is carved before its right end.

Print the pattern.

Input

The only line contains one integer n ($1 \le n \le 1000$): the number of rows.

Output

Print n lines, the rows of the pattern from top to bottom. Each line must be exactly $2n$ characters long.

Examples

input

```
2
```

output

```
A  0
B1C2
```

input

```
3
```

output

```
A    0
B1  C2
D3E4F5
```

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